Lecture 12

Stabilizer codes

Objectives. *Lecture 11 gave the exact condition for correcting errors, a matrix of numbers for every pair of errors. This lecture asks:* how do commuting Paulis define a code, and which errors does it correct? *A stabilizer code is the space fixed by commuting Pauli operators; with r independent generators on n qubits it has dimension 2^{n-r} (Theorem 12.3). Its syndromes are bits, read off from commutation, and a set of Pauli errors is correctable if every $E_j^\dagger E_k$ either has a nonzero syndrome or acts as a scalar on the code (Theorem 12.6, the main result). Steane's code is the worked example:* $[[7,1,3]]$.

By the end of this lecture you will be able to:

- Multiply Pauli operators with their phases, decide whether two commute, and compute the syndrome of a Pauli error.
- Prove that a stabilizer group with r independent generators on n qubits has a code of dimension 2^{n-r}.
- Prove that Pauli errors with no $E_j^\dagger E_k$ in $N(S) \setminus \langle i\mathbb{1}\rangle S$ are correctable, and that a code of distance $2t+1$ corrects every error on t qubits.
- Verify from its six generators that Steane's code is $[[7,1,3]]$.

12.1 The Pauli group

Lectures 10 and 11 used products such as Z_1Z_2, $X_1\cdots X_6$ and $Y_i = iX_iZ_i$. With phases allowed, they form a group.

Definition 12.1 (The Pauli group)**.** The *Pauli group* $\mathcal{G}_n$ consists of the matrices $cP_1\otimes\cdots\otimes P_n$ with $c\in\{\pm1,\pm i\}$ and each $P_q\in\{\mathbb{1},X,Y,Z\}$. The *weight* $\mathrm{wt}(E)$ of such an E is the number of q with $P_q\neq\mathbb{1}$. We write $P_1\cdots P_n$ for the product without its phase: $IIXZ$ is X_3Z_4, of weight 2.

By $XY=iZ$, $YZ=iX$, $ZX=iY$ and $YX=-iZ$ (the Pauli relations $\sigma_a\sigma_b=\delta_{ab}\mathbb{1}+i\varepsilon_{abc}\sigma_c$), the product of two elements of $\mathcal{G}_n$ is in $\mathcal{G}_n$, computed qubit by qubit; the phases $\pm1,\pm i$ are what make it closed. Two of $\mathbb{1},X,Y,Z$ commute if one is $\mathbb{1}$ or they are equal, and anticommute otherwise. So any $E,F\in\mathcal{G}_n$ commute or anticommute:

$$EF=(-1)^mFE,$$

where m counts the qubits on which both factors are not $\mathbb{1}$ and differ. For instance $(X\otimes Z)(Z\otimes X)=(XZ)\otimes(ZX)=(-iY)\otimes(iY)=Y\otimes Y$, and $X\otimes Z$ and $Z\otimes X$ commute, as $m=2$. Each Pauli string $P_1\otimes\cdots\otimes P_n$ is self-adjoint and squares to $\mathbb{1}$, so $(cP)^2=c^2\mathbb{1}$: an element of $\mathcal{G}_n$ squares to $\mathbb{1}$ if its phase is ±1, and then it is self-adjoint, and to $-\mathbb{1}$ if its phase is $\pm i$.

12.2 Stabilizers, code space, dimension

Definition 12.2 (Stabilizer code). A *stabilizer group* is a subgroup S of $\mathcal{G}_n$ whose elements commute and which does not contain $-\mathbb{1}$. Generators $g_1, \dots, g_r$ of S (every element is a product of them) are *independent* if none is a product of the others. The *stabilizer code* $C(S)$ is the set of vectors ψ with $g\psi = \psi$ for every $g \in S$; Π_C is its projector.

A vector fixed by the generators is fixed by their products, so $C(S)$ is the common +1 eigenspace of $g_1, \dots, g_r$. As $-\mathbb{1} \notin S$, no element of S has the phase $\pm i$, since its square would be $-\mathbb{1}$: every $g \in S$ has $g^2 = \mathbb{1}$ and is self-adjoint, and $\pm i\mathbb{1} \notin S$. The eight observables of Shor's code (§10.4) generate a stabilizer group: they commute, and a product of some of them that uses an X-product has an X or a Y somewhere (the two X-products multiply to $X_1X_2X_3X_7X_8X_9$), and one that uses only Z-pairs has a Z (the two pairs of a block multiply to Z_1Z_3, and so on); so it is $\pm\mathbb{1}$ only if it uses none.

Theorem 12.3 (Dimension). *Let $g_1, \dots, g_r$ be independent generators of a stabilizer group $S \subseteq \mathcal{G}_n$. Then*

$$\Pi_C = \textstyle\prod_{j=1}^{r} \frac{1}{2}(\mathbb{1} + g_j),$$

and $C(S)$ has dimension 2^{n-r}.

Proof. Each $\frac{1}{2}(\mathbb{1} + g_j)$ is the projector onto the +1 eigenspace of g_j, as g_j is self-adjoint with $g_j^2 = \mathbb{1}$, and they commute; so their product is the projector onto the common +1 eigenspace, which is $C(S)$. Expanding the product,

$$\Pi_C = 2^{-r} \textstyle\sum_{a \in \{0,1\}^r} g_1^{a_1} \cdots g_r^{a_r}.$$

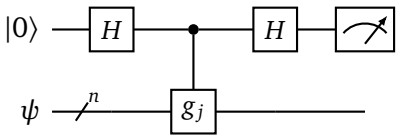
For $a \neq 0$, $g^a = g_1^{a_1} \cdots g_r^{a_r}$ is in S, so it is $\pm P$ for a Pauli string P. If P were $\mathbb{1}$, then g^a would be $-\mathbb{1}$, which S does not contain, or $\mathbb{1}$, which makes each g_j with $a_j = 1$ the product of the others, as $g_j^{-1} = g_j$. So $P \neq \mathbb{1}$, and $\operatorname{Tr} P = \prod_q \operatorname{Tr} P_q = 0$, as $\operatorname{Tr} X = \operatorname{Tr} Y = \operatorname{Tr} Z = 0$. Hence $\dim C(S) = \operatorname{Tr} \Pi_C = 2^{-r} \operatorname{Tr} \mathbb{1} = 2^{n-r}$. ∎

So n qubits and r independent generators encode $k = n - r$ qubits; for Shor's code $n = 9$, $r = 8$ and $k = 1$, the plane of $|0_L\rangle$ and $|1_L\rangle$. Both hypotheses bite. $\langle Z, -Z\rangle$ on one qubit contains $-\mathbb{1}$ and fixes no vector but 0, and the dependent Z_1Z_2, Z_2Z_3, Z_1Z_3 fix the plane of the bit-flip code, of dimension 2, not 2^{3-3}. The proof is Watrous's (2025, pp. 420–421); Nielsen and Chuang prove it otherwise (Proposition 10.5, p. 458).

12.3 Syndromes

An error $E \in \mathcal{G}_n$ commutes or anticommutes with each generator. Its *syndrome* is $s(E) \in \{0,1\}^r$ with $s_j = 0$ if E commutes with g_j and $s_j = 1$ if it anticommutes (Gottesman, (3.9), p. 20; Nielsen and Chuang, p. 466, use the eigenvalues $(-1)^{s_j}$). As in §10.4, for ψ in the code $g_jE\psi = (-1)^{s_j}Eg_j\psi = (-1)^{s_j}E\psi$, so $E\psi$ lies in the range of $\Pi_s = \prod_j \frac{1}{2}(\mathbb{1} + (-1)^{s_j}g_j)$, and the Π_s form the syndrome measurement. Signs multiply, so syndromes add: $s(EF) = s(E) + s(F)$ modulo 2. And $s(E) = 0$ exactly when E commutes with every generator, so with all of S: when E is in the *normaliser* $N(S)$, the set of elements of $\mathcal{G}_n$ that commute with every element of S (as $-\mathbb{1} \notin S$, these are the E with $ESE^\dagger = S$).

Each bit is read with an ancilla:



As $H|0\rangle = |+\rangle$, the controlled g_j and the second H give $|0\rangle \otimes \frac{1}{2}(\mathbb{1} + g_j)\psi + |1\rangle \otimes \frac{1}{2}(\mathbb{1} - g_j)\psi$. By (3.2), the reading gives b with probability $\left\|\frac{1}{2}(\mathbb{1} + (-1)^b g_j)\psi\right\|^2$ and leaves the register in that vector, normalised: the measurement of the eigenvalue $(-1)^b$ of g_j. For g_j a product of Z's, the controlled g_j is a row of controlled Z's, and the circuit can be traded for Lecture 10's CNOTs into the ancilla.

12.4 Distance; correctability

Write $\langle i\mathbb{1}\rangle S$ for the set of cg with $c \in \{\pm 1, \pm i\}$ and $g \in S$.

Lemma 12.4 (Paulis on the code). *Let S have independent generators, $C = C(S)$, and $E \in \mathcal{G}_n$.*

(i) If $E \notin N(S)$, then $\Pi_C E \Pi_C = 0$.

(ii) If $E = cg$ with $g \in S$, then $E\Pi_C = c\Pi_C$.

(iii) If $E \in N(S) \setminus \langle i\mathbb{1}\rangle S$, then E maps C into itself, and is not a multiple of $\mathbb{1}$ on C.

Proof. (i) Some $g \in S$ anticommutes with E, and $g\Pi_C = \Pi_C = \Pi_C g$, as g is self-adjoint and fixes C. So $\Pi_C E \Pi_C = \Pi_C g E \Pi_C = -\Pi_C E g \Pi_C = -\Pi_C E \Pi_C$. (ii) $g\Pi_C = \Pi_C$.

(iii) For $\psi \in C$ and $g \in S$, $gE\psi = Eg\psi = E\psi$, so $E\psi \in C$. Suppose E acts as a scalar on C, so $E\Pi_C = c\Pi_C$. As $E^2 = \pm\mathbb{1}$ and $C \neq 0$, $c^2 = \pm 1$. The operator $F = c^{-1}E$ is in $\mathcal{G}_n$, fixes every vector of C and commutes with S. It is not in S, since $E \notin \langle i\mathbb{1}\rangle S$. Also $F^2 = \pm\mathbb{1}$ fixes C, so $F^2 = \mathbb{1}$.

Adjoin F to the generators. The resulting group S' has elements g and gF with $g \in S$, which commute. It does not contain $-\mathbb{1}$: $gF = -\mathbb{1}$ would give $F = -g$, and then $\psi = F\psi = -g\psi = -\psi$ for $\psi \in C$, contradicting $C \neq 0$. The $r+1$ generators are independent because $F \notin S$.

Compare the dimensions. By Theorem 12.3, the space $C(S')$ has dimension 2^{n-r-1}. But $C(S') = C$, since F fixes C, and C has dimension 2^{n-r}. This contradiction proves that E is not a multiple of $\mathbb{1}$ on C. ∎

The elements of $N(S) \setminus \langle i\mathbb{1}\rangle S$ are the *logical errors*: invisible to the syndrome, and harmful on the code.

Definition 12.5 (Distance). The *distance* d of a stabilizer code is the least weight of an element of $N(S) \setminus \langle i\mathbb{1}\rangle S$, if it has one. A code of n qubits, dimension 2^k and distance d is an $[[n, k, d]]$ code.

Nielsen and Chuang (p. 467) take the least weight in $N(S) - S$, which contains $-\mathbb{1}$, of weight 0; removing the phases ± 1, $\pm i$, as $\langle i\mathbb{1}\rangle S$ does, mends it.

Theorem 12.6 (Correctable Pauli errors). *Let S have independent generators, and let $E_1, \ldots, E_m \in \mathcal{G}_n$. If no $E_j^\dagger E_k$ lies in $N(S) \setminus \langle i\mathbb{1}\rangle S$, then some recovery corrects $E_1, \ldots, E_m$ on $C(S)$.*

Proof. Each $E_j^\dagger E_k$ is in $\mathcal{G}_n$. If it is not in $N(S)$, then $\Pi_C E_j^\dagger E_k \Pi_C = 0$ by Lemma 12.4(i); otherwise it is in $\langle i\mathbb{1}\rangle S$, and $\Pi_C E_j^\dagger E_k \Pi_C = c\Pi_C$ by (ii). So the condition of Theorem 11.2 holds. ∎

Two errors may share a syndrome: $E_j^\dagger E_k \in S$ is allowed, as for Z_1 and Z_2 on Shor's code, where $Z_1 Z_2 \in S$. The code is then degenerate (§11.6), and the recovery need not tell the two apart. What is forbidden is a pair that differs by a logical error. Nielsen and Chuang's Theorem 10.8 (p. 466) asks $E_j^\dagger E_k \notin N(S) - S$, which also excludes pairs such as X_1 and $-X_1$, whose product $-\mathbb{1}$ meets the condition of Theorem 11.2.

> **Corollary 12.7** (Distance $2t+1$)**.** *If $d \geq 2t+1$, some recovery corrects every element of $\mathcal{G}_n$ of weight at most t, and so undoes every noise whose Kraus operators each act on at most t qubits.*

Proof. For E and F of weight at most t, $E^\dagger F$ has weight at most $2t < d$, so it is not in $N(S) \setminus \langle i\mathbb{1}\rangle S$; Theorem 12.6 applies to these finitely many errors. A matrix on t qubits is a combination of the 4^t products of $\mathbb{1}, X, Y, Z$ on them, so Theorem 11.5 gives the rest. ∎

For general codes Kitaev, Shen and Vyalyi prove the analogue with their distance (Proposition 15.4, p. 160).

12.5 Steane's code

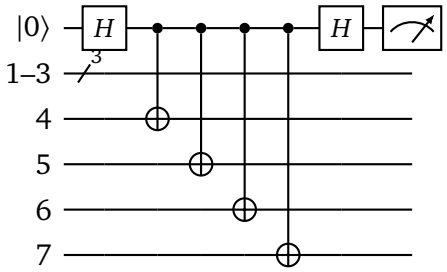
Steane's code has seven qubits and the six generators (Nielsen and Chuang, Fig. 10.6, p. 456, qubit 1 on the left)

$$
\begin{aligned}
g_1 &= IIIXXXX,\\
g_2 &= IXXIIXX,\\
g_3 &= XIXIXIX,\\
g_4 &= IIIZZZZ,\\
g_5 &= IZZIIZZ,\\
g_6 &= ZIZIZIZ.
\end{aligned}
$$

Qubit i carries an X in g_1, g_2, g_3, and a Z in g_4, g_5, g_6, according to the binary digits of i, the first generator for the digit 4. An X-type and a Z-type generator share 2 or 4 qubits, so all six commute. Qubits 4, 2 and 1 carry an X in g_1, g_2, g_3 alone and a Z in g_4, g_5, g_6 alone, so a nonempty product has an X, Y or Z on one of them, as $XZ = -iY$. So no nonempty product is $\pm\mathbb{1}$: the generators are independent and $-\mathbb{1} \notin S$. By Theorem 12.3 the code has dimension $2^{7-6} = 2$: $k = 1$.

Syndromes. Z_i anticommutes with the X-type generators with an X on qubit i: its syndrome is i in binary, then 000. X_i gives 000, then i in binary, and $Y_i = iX_iZ_i$ both. So the 21 one-qubit Paulis have distinct, nonzero syndromes. Each bit is read by §12.3's circuit; for

$g_1 = X_4X_5X_6X_7$ the controlled g_1 is four CNOTs:



Distance. A Pauli of weight 1 has a nonzero syndrome. A Pauli of weight 2, A_iB_j with $i \neq j$, has the syndrome $s(A_i)+s(B_j) \neq 0$, as the 21 syndromes are distinct. So $N(S)$ has no element of weight 1 or 2, and $d \geq 3$. And $X_1X_2X_3 = XXXIIII$ is in $N(S) \setminus \langle i\mathbb{1}\rangle S$: it shares qubits 2, 3 with g_5, 1, 3 with g_6 and none with g_4, so it commutes with all six; and the elements of S whose strings have only I and X are $\mathbb{1}$ and the seven nonempty products of g_1, g_2, g_3, each of weight 4. So $d = 3$: Steane's code is $[[7, 1, 3]]$, and by Corollary 12.7 it corrects every one-qubit error, with seven qubits where Shor's code used nine. Watrous (2025, pp. 418 and 427) lists the same generators little-endian, each string reversed; read in our order, his strings give a code in which XXXIIII is detected and IIIIXXX is the logical error.

Summary.

- *The Pauli group:* phases $\pm 1, \pm i$; two elements commute or anticommute, by the parity of the qubits where both are not $\mathbb{1}$ and differ; weight.
- *Stabilizer codes:* r independent commuting generators with $-\mathbb{1} \notin S$ fix a code of dimension 2^{n-r} (Theorem 12.3).
- *Syndromes:* bits by anticommutation; they add, and vanish exactly on $N(S)$.
- *The main result:* if no $E_j^\dagger E_k$ lies in $N(S) \setminus \langle i\mathbb{1}\rangle S$, the Pauli errors are correctable (Theorem 12.6); distance $2t + 1$ corrects every error on t qubits (Corollary 12.7).
- *Steane:* $[[7, 1, 3]]$, from its six generators.
- *Next:* Lecture 13, Clifford circuits.

References

Nielsen and Chuang, pp. 454–458 and 465–467. Watrous (2025), pp. 403, 414–428 (little-endian). D. Gottesman, arXiv:quant-ph/9705052 (1997), pp. 15 and 20. Kitaev, Shen and Vyalyi, pp. 160 and 169.